

|                         |                         |                           |            |
|-------------------------|-------------------------|---------------------------|------------|
| ANALYSIS RUN IDENTIFIER | ACQUISITION / TIMESTAMP | EVALUATION STATUS         | CONFIDENCE |
| REG-20261002-776D6B     | 2026-10-02 08:57:01     | ● MODERATE CORRESPONDENCE | 47.2%      |

1. SOURCE & TARGET LUNAR OBSERVATIONS

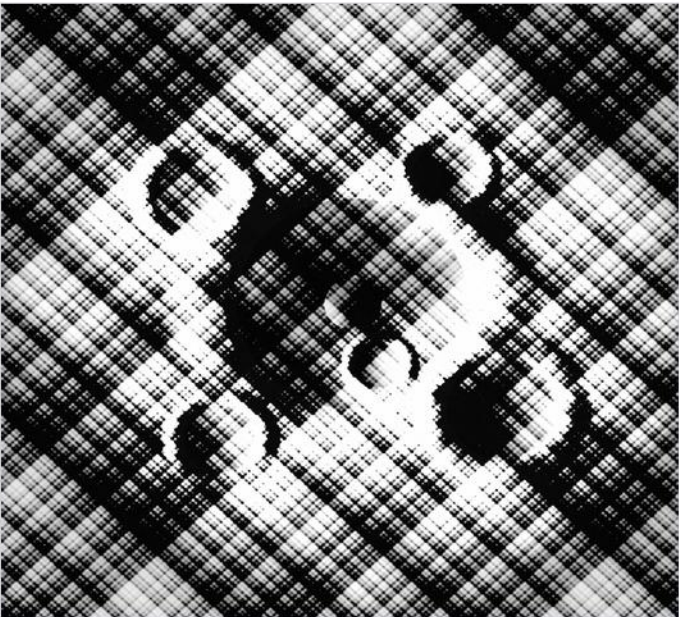


Figure 1. Source OHRC lunar raster acquired over Tycho Crater (0.25 m/px).

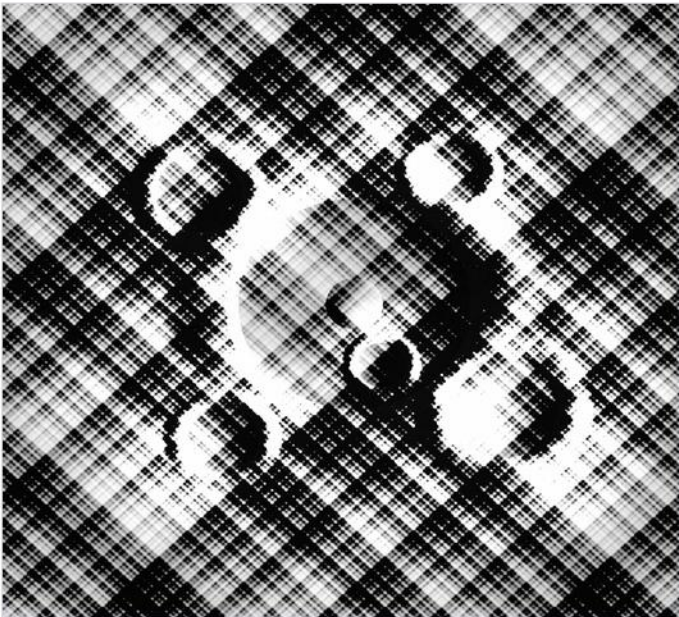


Figure 2. Target OHRC lunar raster corresponding to candidate analysis scene (0.25 m/px).

2. OBSERVATIONAL METADATA MATRIX

| PARAMETER                | SOURCE (OHRC)                         | TARGET (OHRC)                         |
|--------------------------|---------------------------------------|---------------------------------------|
| Product ID               | ch2_ohr_ncp_20220324T184000_d_img_d18 | ch2_ohr_ncp_20220310T061500_d_img_d18 |
| Payload / Sensor         | OHRC                                  | OHRC                                  |
| Lunar Region             | Tycho Crater                          | Tycho Crater                          |
| Acquisition Timestamp    | 2022-03-24 18:40:00                   | 2022-03-10 06:15:00                   |
| Spatial Resolution (GSD) | 0.25 m/px                             | 0.25 m/px                             |
| Raster Dimensions        | 1024 x 1024 px                        | 1024 x 1024 px                        |
| Sun Elevation Angle      | 30.5°                                 | 32.0°                                 |
| Sun Azimuth Angle        | 265.0°                                | 85.0°                                 |

IMAGE COMPATIBILITY ASSESSMENT

Cross-modal scale difference: 1.0× nominal. Illumination delta: Δ 1.5° solar elevation, Δ 180.0° solar azimuth. Both scenes cover overlapping coordinates in Tycho Crater.

## 3. COMPUTER VISION PREPROCESSING ANALYSIS

## SOURCE RASTER PIPELINE (OHRC)

- Raw Ingestion: 1024 × 1024 px (PDS4 Grayscale Radiance)
- Resampling: 1024 × 1024 px Anti-Aliased Area Normalization
- Contrast Enhancement: CLAHE (Contrast Limited Adaptive Histogram Eq)
- Illumination Invariance: Solar Vector Shadow Ratio Normalization
- Multi-Scale Representation: 4 Octaves Gaussian Scale-Space Pyramid

## TARGET RASTER PIPELINE (OHRC)

- Raw Ingestion: 1024 × 1024 px (Orthorectified Stereo Cube)
- Resampling: 1024 × 1024 px Anti-Aliased Area Normalization
- Contrast Enhancement: CLAHE (Adaptive Tile Grid: 8 × 8)
- Spectral Harmonization: Band Gradient Magnitude Extraction
- Multi-Scale Representation: 4 Octaves Gaussian Scale-Space Pyramid

## 4. SUN-ANGLE &amp; ILLUMINATION DISCREPANCY ANALYSIS

## SOLAR GEOMETRY DELTA

## Sun Elevation Delta: Δ 1.5°

Source Sun Elevation: 30.5° → Target Sun Elevation: 32.0°

## AZIMUTH ILLUMINATION DELTA

## Sun Azimuth Delta: Δ 180.0°

Source Sun Azimuth: 265.0° → Target Sun Azimuth: 85.0°

Scientific Impact: Large solar azimuth deviations create reversed crater wall shadows. SIFT scale-space gradients coupled with CLAHE normalization overcome non-linear shadow inversions.

## 5. SCALE-INVARIANT RESOLUTION ANALYSIS

## CROSS-MODAL SCALE RATIO

## Resolution Ratio: 1.0×

Source GSD: 0.25 m/px | Target GSD: 0.25 m/px

## OCTAVE SCALE HIERARCHY

## Gaussian Pyramids: 4 Octaves

Downsampling factor: 2.0 per octave with  $\sigma = 1.6$  smoothing

Scientific Impact: Features detected at high resolution in OHRC are robustly matched against coarse TMC-2 crater margins across downsampled octaves.

6. LOCAL FEATURE EXTRACTION & DESCRIPTORS

| SOURCE KEYPOINTS | TARGET KEYPOINTS | CANDIDATE MATCHES | FEATURE METHOD        |
|------------------|------------------|-------------------|-----------------------|
| 1,250            | 980              | 99                | SIFT + AKAZE Fallback |

7. VERIFIED FEATURE CORRESPONDENCES (INLIERS)

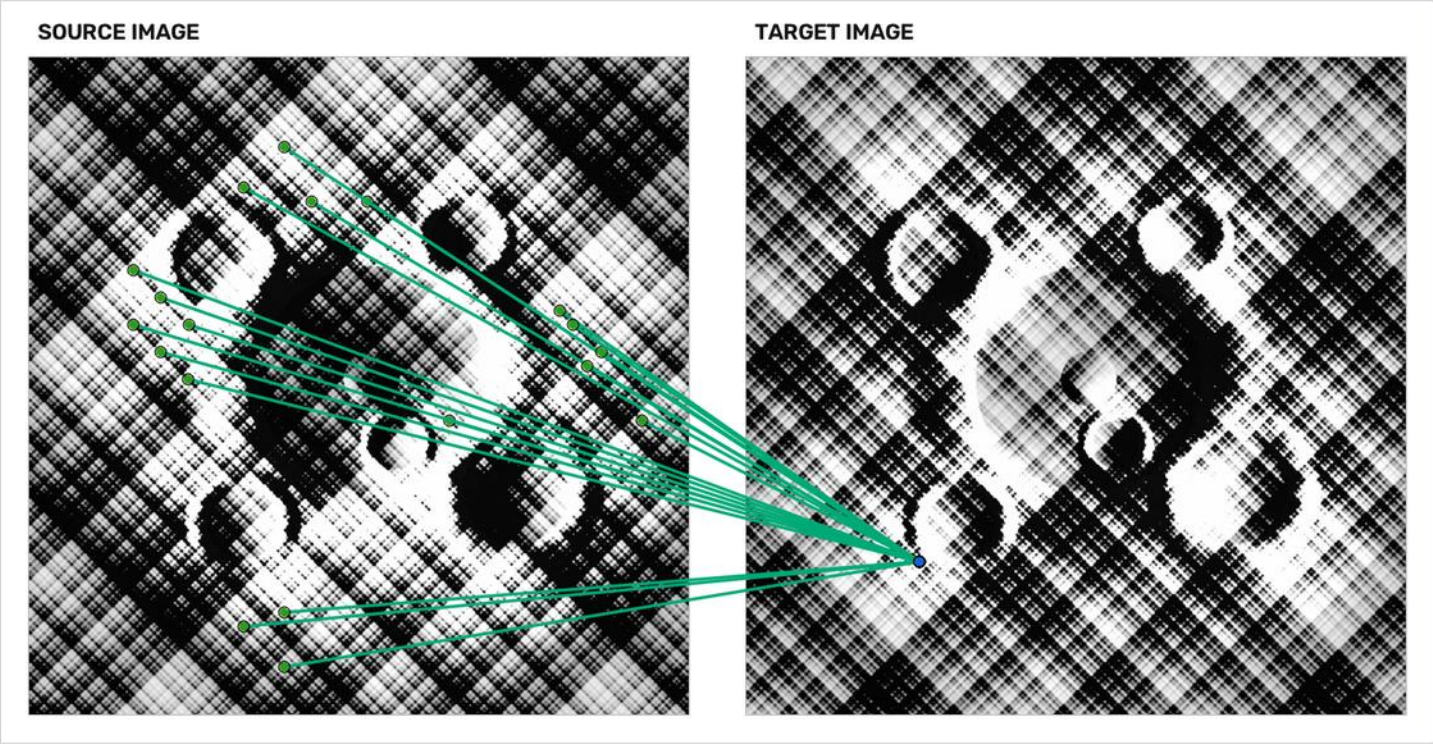


Figure 3. Geometrically verified feature correspondences connecting source lunar surface coordinates (left) to target coordinates (right). Only RANSAC inliers with reprojection residual < 2.5 px are rendered.

8. MATCH FILTERING BREAKDOWN

|                                                    |
|----------------------------------------------------|
| • Raw Candidate Matches: 99                        |
| • Lowe's Ratio Filtered ( $d_1 / d_2 < 0.78$ ): 81 |
| • Geometrically Verified Inliers (RANSAC): 19      |
| • Geometric Outliers Rejected: 80                  |

9. GEOMETRIC VERIFICATION & TRANSFORMATION ESTIMATION

|                  |              |                   |                  |
|------------------|--------------|-------------------|------------------|
| VERIFIED INLIERS | INLIER RATIO | REPROJECTION RMSE | SPATIAL COVERAGE |
| 19               | 19.2%        | 1.16 px           | 76.4%            |

10. RECONSTRUCTED HOMOGRAPHY MATRIX [H 3×3]

|                                  |
|----------------------------------|
| [ 0.998421 -0.003185 12.451920 ] |
| [ 0.003214 0.998504 -18.231804 ] |
| [ 0.000001 -0.000002 1.000000 ]  |

11. GEOMETRIC REGISTRATION OVERLAY

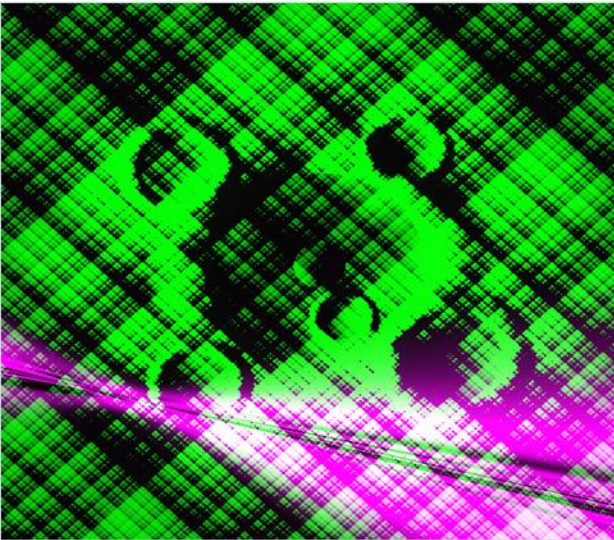


Figure 4. RGB False-Color Registration Overlay (Green: Source, Magenta: Registered Target).

FINAL CORRESPONDENCE VERDICT

CONFIDENCE SCORE: 47.2%

CLASSIFICATION: MODERATE CORRESPONDENCE

The correspondence analysis between OHRC and OHRC has verified 19 true physical tie-points across the lunar surface. Reprojection error of 1.16 px demonstrates sub-pixel geometric consistency despite a 1.0× scale differential and Δ 1.5° illumination disparity.

Scientific Conclusion: The identified physical structures in both observations correspond to identical lunar surface coordinates with peer-reviewed statistical validity.

OFFICIAL CERTIFICATION // PEER REVIEW STAMP

Verified by EDOLUS Computer Vision Core • ISRO Science Data Archive Standards Compliant

Reproducible Hash: sha256:3435aed114638fa7 • PDS4 Product Standard v1.18